Technical Proposal

Design and Development of Campus Development Office Workflow Digitalization and Automation Platform

for

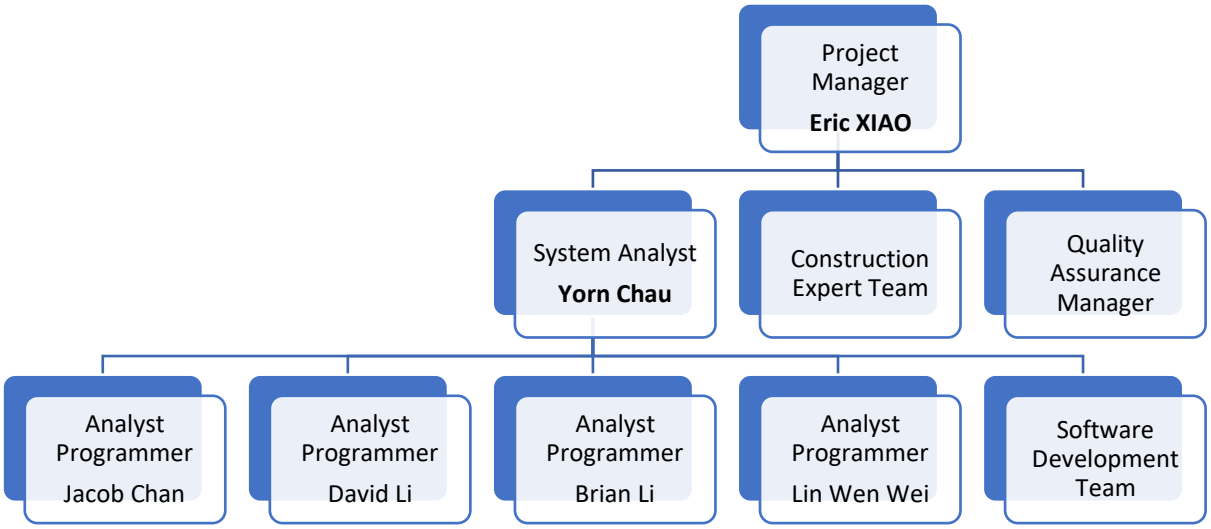
The Hong Kong Polytechnic University

Submitted by: isBIM Limited

Date: 29 May 2025

1 Staffing & Qualification

1.1 Organization Chart



1.2 Core Team Profiles *(Attach full CVs in Appendix D)*

Role	Staff Name	Qualification	Years Experience	Responsible Areas and Duties with respect to the work assignment
Project Manager	Eric Xiao	Master of Business Systems	▪ 16	▪ Responsible for the overall management of the project ▪ Schedule and assign duties to subordinates ▪ Lead feasibility and optimisation studies, analyse and specify system ▪ Plan and control all phases of computer project including feasibility study, system analysis and design, procurement and installation of computer systems, programming, implementation, system nursing and maintenance ▪ Deliver projects within budget and on time ▪ Establish procedures and automated performance measurement capability to monitor the technical accomplishment and progress of projects

Role	Staff Name	Qualification	Years Experience	Responsible Areas and Duties with respect to the work assignment
				▪ Liaise with Programme Manager and Government project team and management team on day-to-day work ▪ Address all project issues proactively and promptly, especially on resources allocation and compliance within project schedule ▪ Issues and queries raised and considered as urgent by Client.
System Analyst 1	Yorn Chau	Bachlor of Telecom-munication Engineering	▪ 6	▪ Play the full role in all phases of the project including requirement and feasibility studies, system analysis and design, procurement and installation of computer system, programming, implementation, system nursing and maintenance ▪ Provide supervisory, technical and administrative direction for personnel performing system implementation tasks ▪ Analyse, design and specify program components of systems to meet user requirements ▪ Prepare and conduct system testing and user acceptance ▪ Monitor the performance of subordinates ▪ Review work for rectification and for adherence to the design concepts and user requirements, and for all progress in accordance with schedules ▪ Assure conformance to standards and quality control procedures

Role	Staff Name	Qualification	Years Experience	Responsible Areas and Duties with respect to the work assignment
Analyst Programmer 1	Jacob Chan	Bachlor of Computer Science	▪ 3	▪ Participate in the development of programs and ensure program testing meets user requirements ▪ Produce and maintain well documented programs that adhere to standards and satisfy user requirements, provide end-user support services in computer related matters ▪ Assist in the system analysis and design ▪ Produce project reports and specifications in accordance with the standards approved by the Client
Analyst Programmer 2	David Li	Bachlor of Computer Science	▪ 3	
Analyst Programmer 3	Brian Li	Bachlor of Computer Science	▪ 3	
Analyst Programmer 4	Lin Wen Wei	Bachlor of Computer Science	▪ 11	

1.3 Manhour Allocation & Availability

No.	Name / Placeholder	Role	Project Initiation	Req. Study & Design	Development & Config	Testing & UAT	Training	Doc. & Report	Nursing & Maint.	Total MD
1	Eric Xiao	Project Manager	4	5	10	10	5	5	10	49
2	Yorn Chau	System Analyst	3	8	40	8	3	3	5	70
3	Jacob Chan	Frontend Developer	–	–	40	20	–	–	10	70
4	David Li	Frontend Developer	–	–	40	20	–	–	10	70
5	Brian Li	Frontend Developer	–	–	40	20	–	–	10	70
6	Lin Wen Wei	Backend Developer	–	5	20	20	5	5	10	65
7	Du Zhi Heng	IT Engineer	2	2	10	10	–	–	10	34
8	Ruby Hung	Business Analyst	5	10	10	6	3	5	–	39
–		Total Man-Days	19	50	270	170	39	28	85	467

2 Tenderer's Experience on Similar Projects (See Appendix for Screenshots)

2.1 Digital PM Systems – Hong Kong Construction Sector

Project 1	iSuper 2.0 for New Extension Railway Projects
Project Description	To set up and Construction Project Management Systems for the captioned project. To customize multi workflow RICS form, Site Observation form, NCR form etc. To integrate with Material Testign System and ACC BIM Platform via API. With a customized dashboard offering automated data analysis, iSuper 2.0 has demonstrably enhanced project traceability and provided valuable insights.
Client	MTR Corporation Limited
Timeline	Develop: 2021-2022; Launch:2022
Contract Sum	HKD $3,000,000+ (Accumulating)
Number of eForms	11 for over 70 projects
Users	2500 (growing 1000 each year)

Project 2	Digital Mobile Platform for AM/FM
Project Description	To streamline maintenance workflows in educational institutions. To allow users to report facility issues instantly via mobile, and automates the entire day-to-day repair procedure from defect submission to resoultion. DMP integrates seamlessly with the AANI platform to synchronize the maintenance record, providing evidence for further commercial process, ensuring effiency, transparceny and collaboration across deparments.
Client	Vocational Training Council
Contract Sum	HKD $1,000,000+ (Accumulating)
Timeline	Develop: 2024-2025; Launch:2025
Number of eForms	14 for over 30 campuses locations
Users	300 internal, 10,000 external

Project 3	Enterprise Project Document Management System for Woon Lee
Project Description	To centralize platform designed to standardize and streamline project-related documentation processed. It provide predefined templates Site Letters, Contractor Submissions, RFIs and Transimittals, eunsuring consistency and compliance across projects. PDMS features contract

	dashboard and portfolio dashboard to monitor roganizational-wide performance metrics.
Client	Woon Lee
Contract Sum	HKD $1,000,000+ (Accumulating)
Timeline	Develop: 2024-2025; Launch:2025
Number of eForms	21 for over 10 projects
Users	300 internal, 10,000 external

2.2 Digital PM Systems – Overseas / non-HK

Project 4	中国人保公司武汉市分公司建工安责险风控平台
Project Description	The Safety Liability Insurance Platform is an information and risk control solution designed for insurance companies. It includes functions such as risk assessment, hazard identification, emergency management, risk control services, and data analysis, providing comprehensive business management capabilities to support safety liability insurance operations.
Client	中国人保公司武汉市分公司
Contract Sum	RMB ¥1,000,000+
Timeline	Develop: 2021-2022; Launch:2022
Number of eForms	25
Users	100

3 Response to the Brief

3.1 Understanding of Objectives

3.1.1 Our Understanding of PolyU's Objectives

PolyU's Campus Development Office (CDO) seeks a single, cloud-ready Core Operation Platform that will:

- **Unify data** for 50 + concurrent capital-works projects in one trusted source of truth.
- **Digitalise and automate** roughly 40 paper-based forms and their approval workflows.
- **Surface real-time insight** on cost, schedule, risk and compliance through role-based dashboards.
- **Lay an extensible foundation** for future AI-driven analytics and seamless integration with other PolyU enterprise systems.

3.1.2 Key Constraints & Issues We Have Addressed

Tender Constraint / Issue	isBIM's Purpose-Built Response

Tight timeline: Phase 1 Core Platform live by **Q3 2025**; Phase 2 workflow automation by **Q4 2025**; full roll-out by year-end 2025.	Six-month Agile delivery plan (Workshop → Configure → UAT → Roll-out) with 2-week sprints, weekly demos and a "working master-job-list" prototype and workflow automation in 3-4 weeks.
Large form set (≈ 40)	Low-code eForm builder converts a one-page A4 form—including workflow and role matrix—into a production eForm **within one day**.
Data migration of 50 live projects; manual back-loading permitted.	Pre-configured Excel/CSV importers + bulk "wizard" minimise manual entry and ensure data quality; optional plug-in scripts accelerate legacy data migration.
PolyU NetID SSO & MFA for internal users; segregated log-ins for external parties.	Native SAML 2.0 SSO, Microsoft Entra ID (AD) ready; external consultants receive granular eForm-only access with audit trails.
Cloud first, but on-prem options for DR and archiving.	Deployed on HK Huawei Cloud clusters with auto-scaling SSD / HDD tiers; optional MinIO-based on-prem cold storage.
ISO 19650 data governance, 24 × 5 support, 3-year enhancement commitment.	isBIM is ISO 9001 & ISO 19650 certified; JARVIS eForm System already approved by MTR and VTC and many construction clients; 24 × 5 help-desk and bi-annual feature packs written into SLA.

3.1.3 Special Requirements Met Out-of-the-Box

- **Master Job List** auto-generates project snapshots and monthly PDF reports.
- **Dashboard twins** – project-level and portfolio-level – driven by live data, not static exports.
- **Built-in retention & purge** policies (admin-triggered archival; 30-day incremental backups).
- **Security Questionnaire compliance** – previous deployments passed OGCIO & MTR security audits; our template answers PolyU's Appendix C in full.

3.1.4 isBIM Is the best candidate for the assignment

Challenges	**isBIM Advantage**
Proven playbooks for university-scale construction management	2,200 + digital construction projects delivered across HK, Mainland; Recent campus-wide platforms for VTC.
Fast form iterations	Micro-service, low-code architecture → ~ 30 % less configuration effort vs. green-field build; eForms in < 24 h.

Deep construction DNA	Independent of any contractor; team includes BIM, QS, site-safety & AI specialists who speak both engineering and IT.
Scalable, future-ready stack	Containerised services, open APIs, modulized for easy add-ons (AI, IoT, GIS, Digital Twin).
Commitment beyond go-live	Three-year roadmap, 24 × 5 help-desk, local HK team for face-to-face support – no offshore outsourcing.

isBIM's JARVIS-powered Core Operation Platform directly targets CDO's pain-points: fragmented data, manual approvals and limited visibility. By combining **domain-specific templates**, a **low-code form engine**, and a **disciplined six-month Agile schedule**, we will deliver measurable gains:

- **> 40 % shorter approval cycles**
- **Single source of truth** for 50 + live projects within 2025
- **Actionable portfolio dashboards** that turn hindsight into foresight

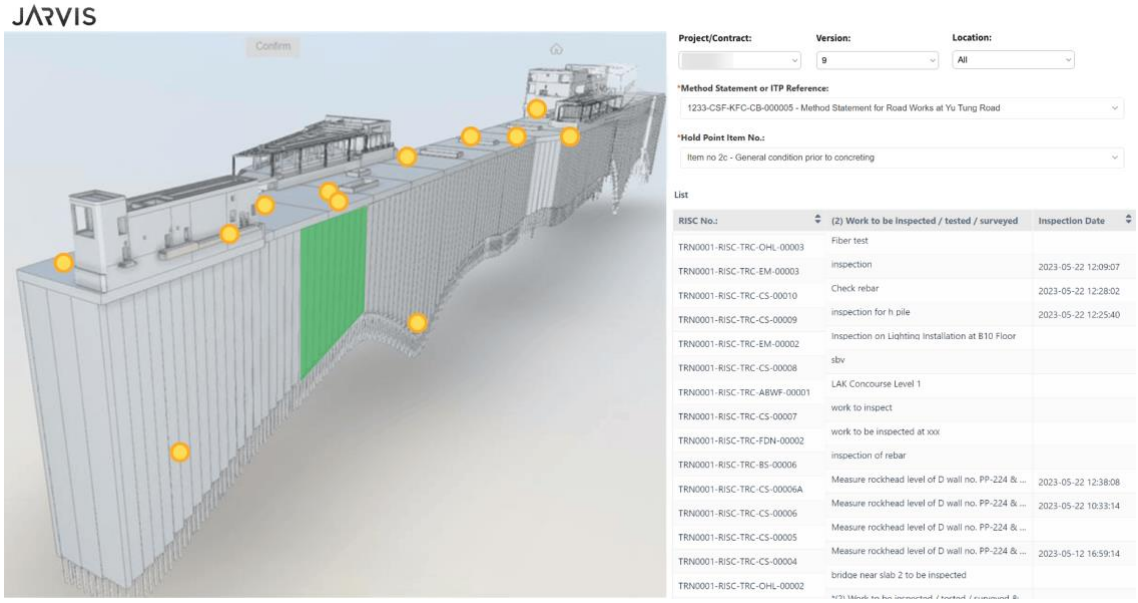
Our regional track record, certified processes, and construction-tech focus make **isBIM the safest, fastest and most future-proof choice** for PolyU's digital-workflow transformation.

3.2 Innovation & Future Expansion

3.2.1 BIM Integration

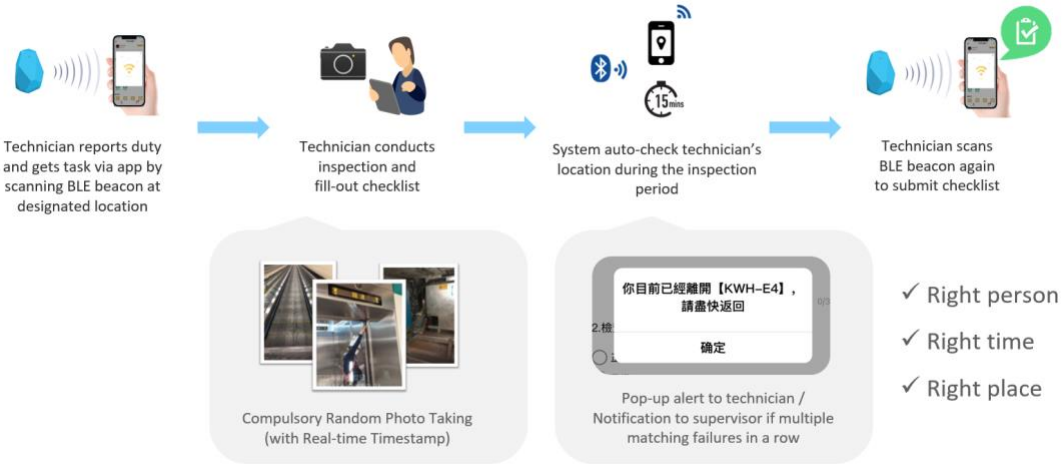
JARVIS DWSS has integrated BIM with DWSS RISC form with our client in MTR. The DWSS can retrieve the relevant BIM data from the BIM objects which are linked to the RISC form.

The RISC form data can also be visualized on the BIM model, the inspected area/objects and defects can be highlighted in different colors.



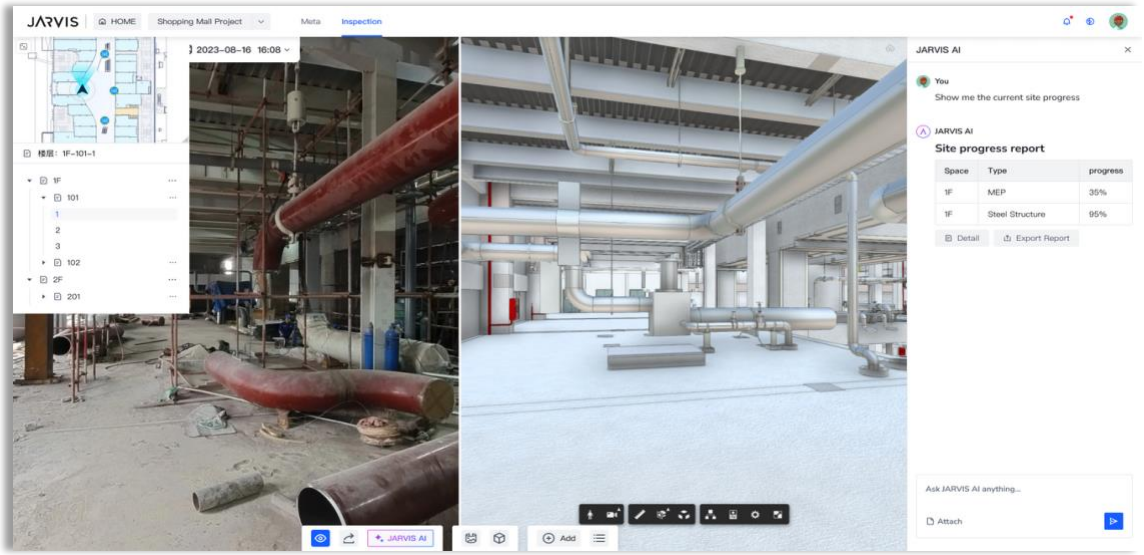
3.2.2 Integration of RFID system

RFID beacons can be placed on site according to the inspection plan. The system will log the inspector's identity, time, and location automatically.



3.2.3 Eagle Eagle Eye Digital Twin Hyper-reality Viewer

JARVIS Eagle Eye is the immersive hyper-reality environment, which not only features the 3D virtual model but also integrate it with the captured environment in the site location. The real image is aligned with 3D model with isBIM's A.I. technology, complements the model with the latest and true reflective of the site environment.



3.2.4 AI Agent with Advanced Analytics Skill:

isBIM's latest DeepSeek/Qwen LLM Driven, AI Agents provide insights into project performance, and analyses historical data for future projects, aiding continuous improvement.

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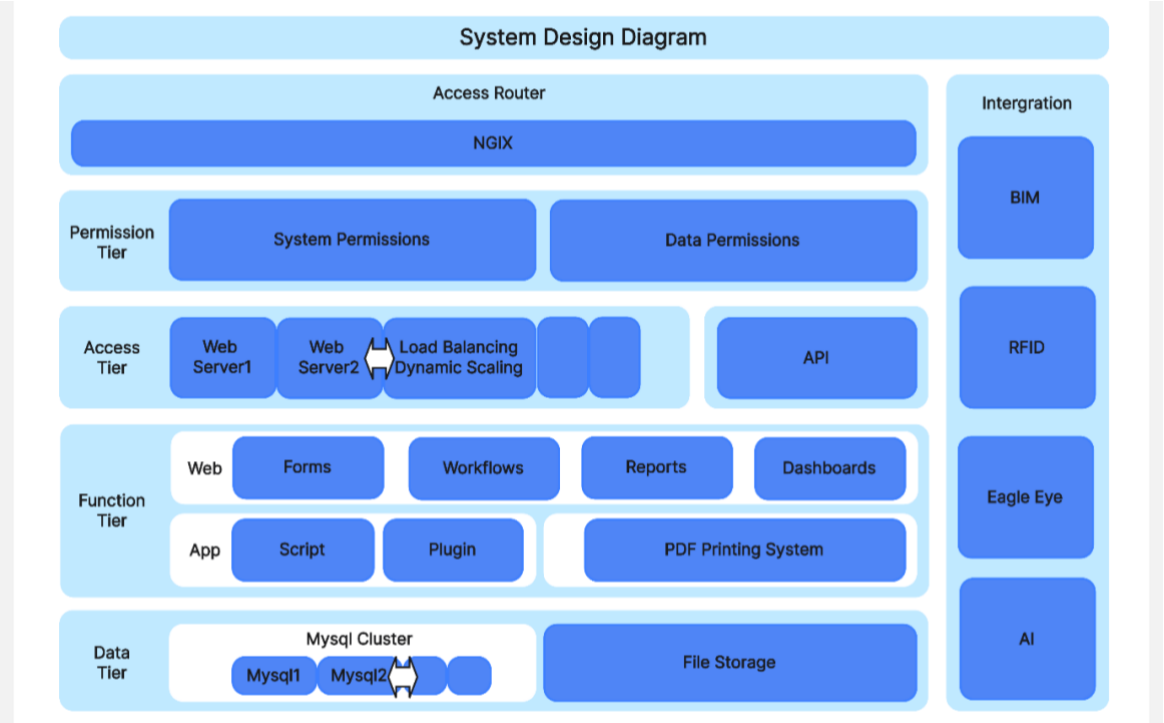
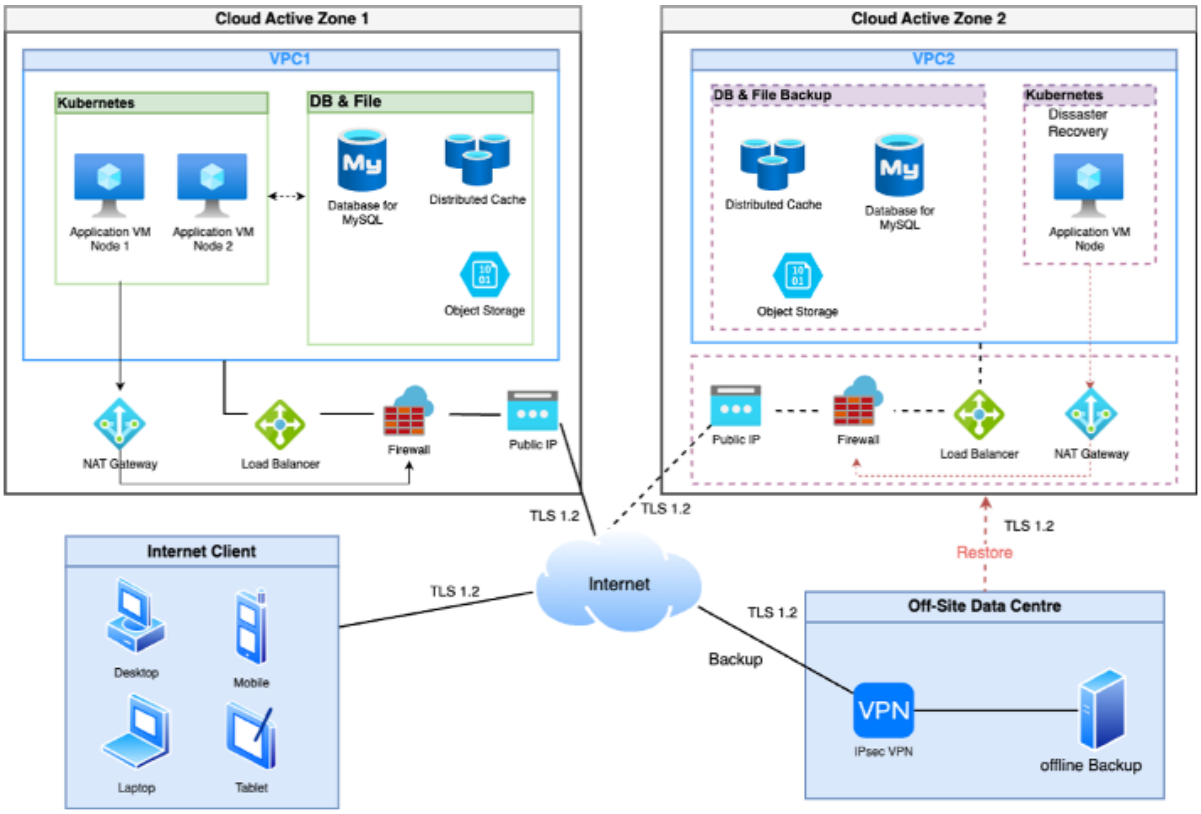
4 Design Approach

4.1 System Design & Data Flow

- **Ingress**: NGIX router receives client requests and routes to Web Servers.
- **Scaling**: Load-balancer auto-scales servers for demand spikes.
- **Permissions Check**: System & data permission layer validates each request.
- **Processing**: APIs trigger functions—forms, workflows, reports, dashboards, scripts, plug-ins, PDF output.
- **Persistence**: Results saved to MySQL cluster and file storage.
- **Integration**: Data/events synchronized with BIM, RFID, Eagle Eye, and AI services.



4.2 System Architecture



4.2.1 System Architecture Design

This system architecture is designed to deliver resilience against failures, ensure data protection, and guarantee business continuity with defined RTO for disaster recovery planning. It showcases a scalable and secure approach to cloud application deployment that can adapt to varying loads while maintaining service continuity and data integrity.

4.2.2 Multiple Nodes for High Availability and Load Management

- **Elastic Compute Service (ECS) Nodes**: The architecture employs two identical servers, Node 1 and Node 2, which run in the daily operational Cloud Active Zone 1 (AZ1). These nodes host the application and database services, ensuring high availability and redundancy.
- **Kubernetes**: The use of Kubernetes orchestrates containerized applications across the two ECS nodes, providing a platform for automated scaling, deployment, and operations of application containers.

4.2.3 Health Check and Load Balancing

- **Server Health Monitoring**: Continuous health checks are conducted to monitor the server health status. This helps in preemptively detecting and addressing issues that might cause downtime or performance degradation.
- **Elastic Load Balancer (ELB)**: The ELB is deployed to manage incoming traffic efficiently. It operates in cluster mode, directing traffic only to healthy servers, thus maintaining service continuity and effective load distribution.

4.2.4 Backup and Data Replication

- **Cross-Zone Backup**: The architecture stipulates daily backups of data and files to a different active zone, Cloud Active Zone 2 (AZ2). This ensures that data is replicated in a geographically separate location to avoid simultaneous loss in case of a localized disaster.
- **Object Storage**: Backed-up data, including databases and files, is stored securely in object storage, which is designed for high durability and availability.

4.2.5 Disaster Recovery

- **Standby Server (ECS Node 3)**: In AZ2, a standby server, Node 3, is ready to be spun up in case of a disaster. It is configured to utilize the backup data from object storage to restore services.
- **Recovery Time Objective (RTO)**: The architecture is designed with an RTO of 4 hours, indicating that the system aims to be restored to operational status within this timeframe following a disaster event.

4.2.6 Security and Connectivity

- **Firewall**: A WAF service is implemented to protect the internal network infrastructure from unauthorized access and various types of cyber threats.
- **VPN and Secure Protocols**: Secure connections are established using VPN and TLS 1.2 protocols, providing encrypted communication channels for data in transit, both within the cloud infrastructure and from external access points, like the on-premises data center.
- **NAT Gateway**: It is used to enable private subnets to connect to the internet or other AWS services but prevent the internet from initiating connections with the instances.

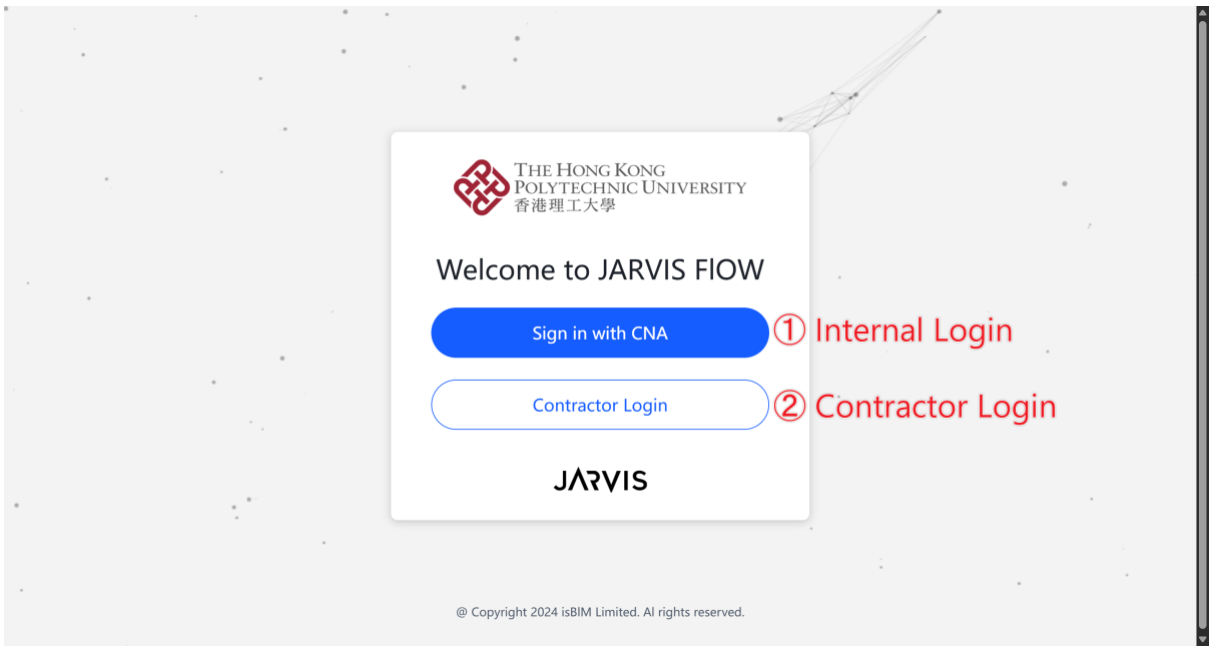
- **SSO Login:** Native SAML 2.0 SSO, Microsoft Entra ID (AD), MFA support for seamless internal user login using NetID.

4.3 Compliance Standards

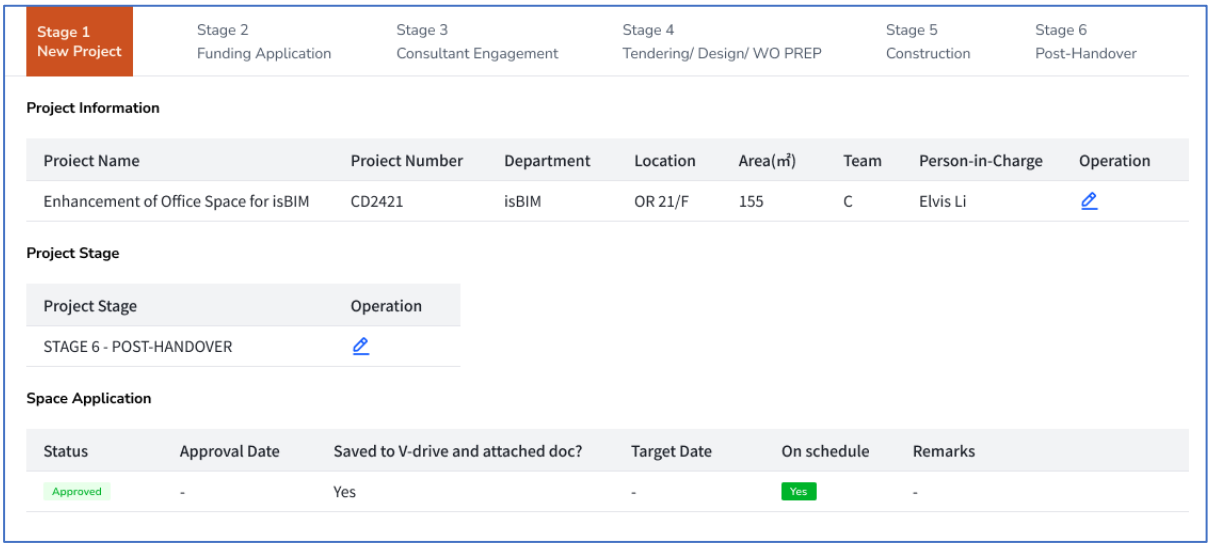
isBIM's solution that meets the DPO's **Baseline IT Security Policy**: (https://www.govcert.gov.hk/doc/G3_EN.pdf) and **The Personal Data (Privacy) Ordinance (**https://www.pcpd.org.hk/english/data_privacy_law/ordinance_at_a_Glance/ordinance.html**)**

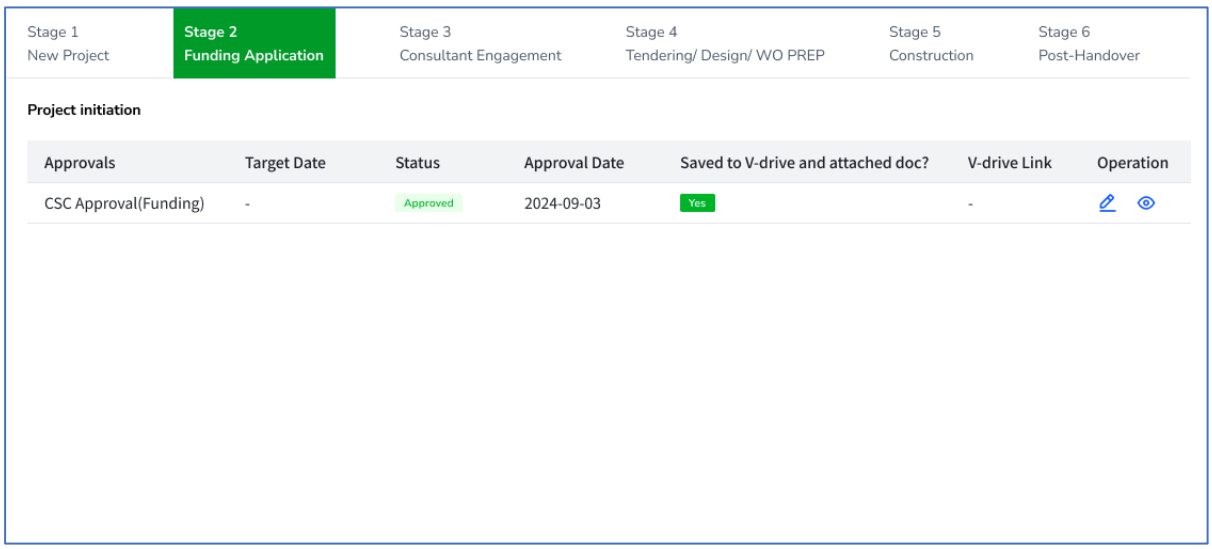
4.4 UI / UX Concept

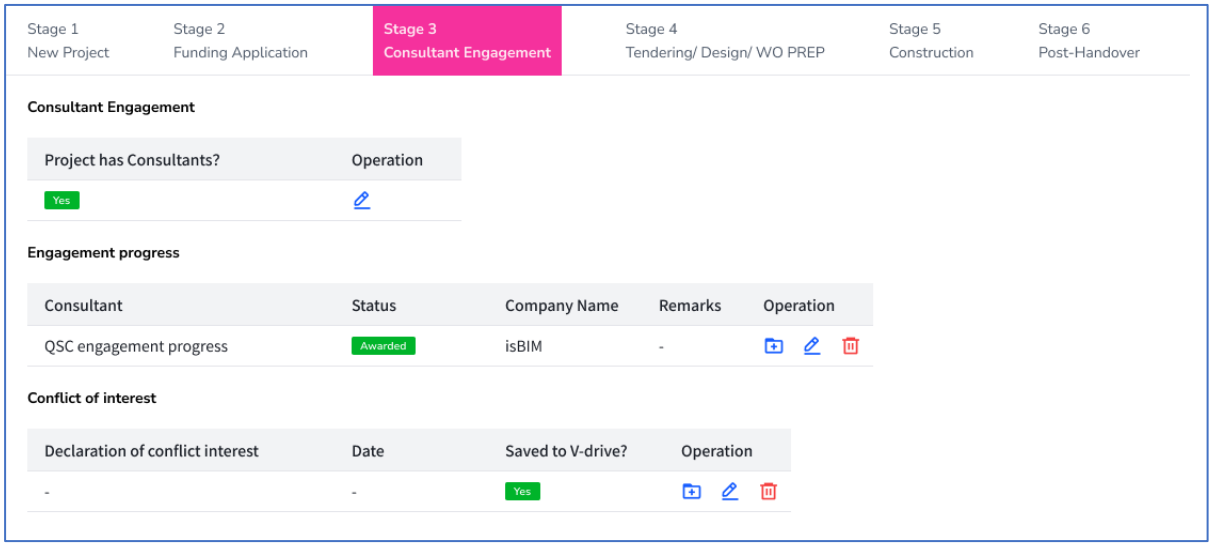
4.4.1 Login: 2 methods for internal login and contractor login



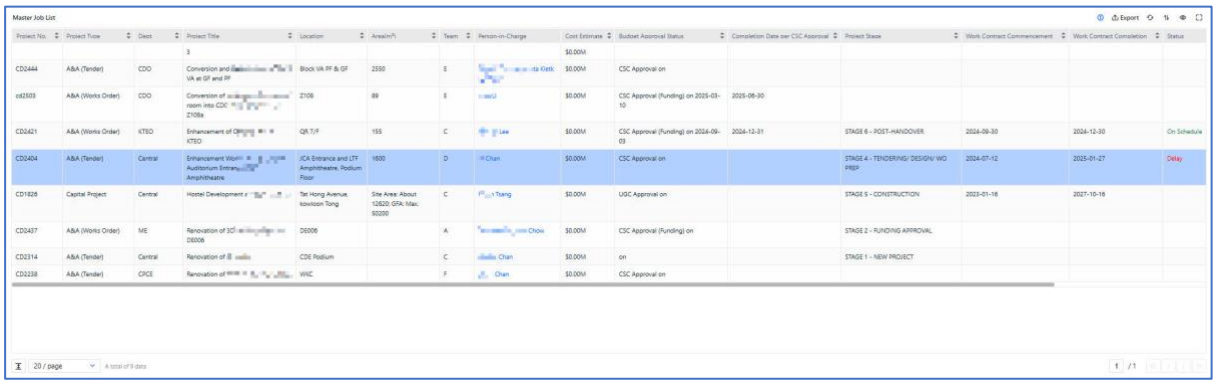
4.4.2 Master Entry Form: Input area with nagvigation bar covering 6 stages of the project.



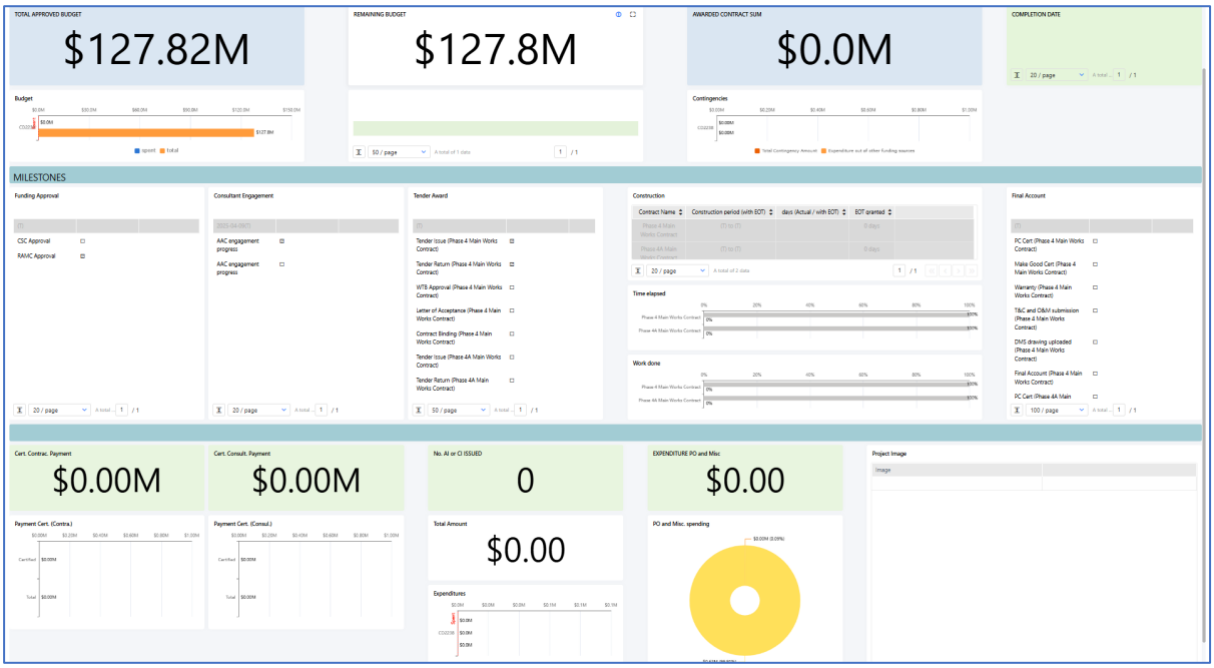




4.4.3 Master Job List: List all the projects and key information with latest updatetime, and could be export as excel.

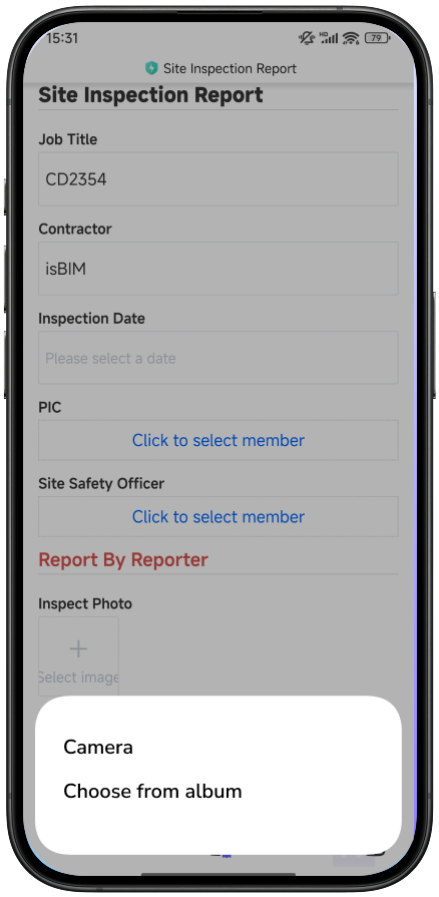


4.4.4 Project Dashboard for example:

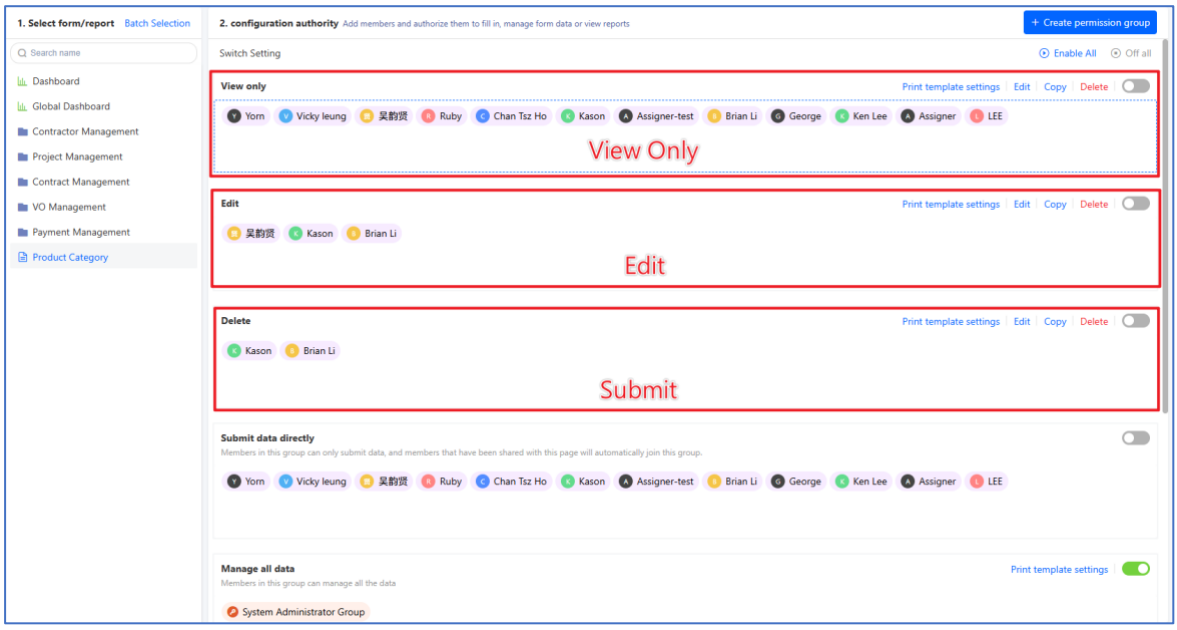


4.4.5 Global Dashboard (overview of all projects) for example

4.4.6 Compatibility on adaptive layouts on phones:

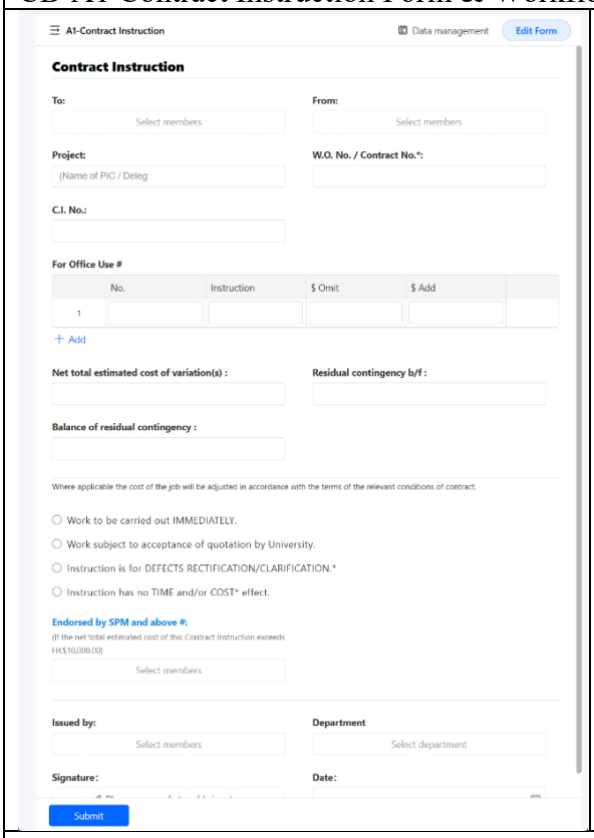
Login on phone	Taking photos in Site Inspeciton Report
	

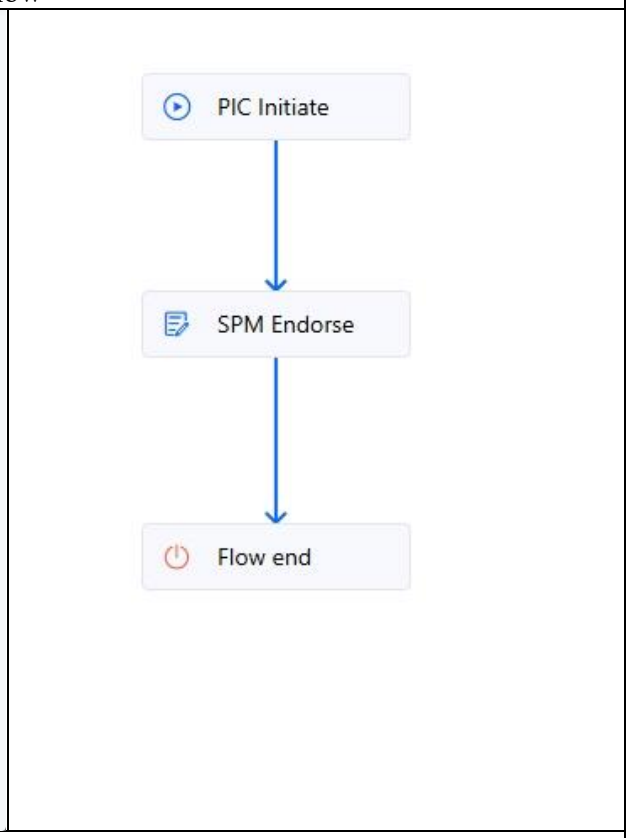
4.4.7 User Management: A module to manage organizations, user accounts, roles of users and permissions.



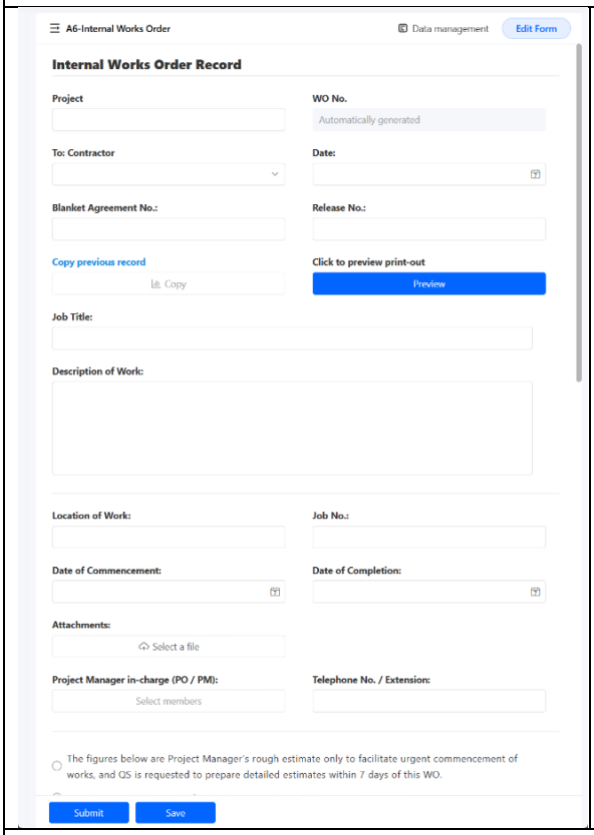
4.4.8 Digitalized Forms for example:

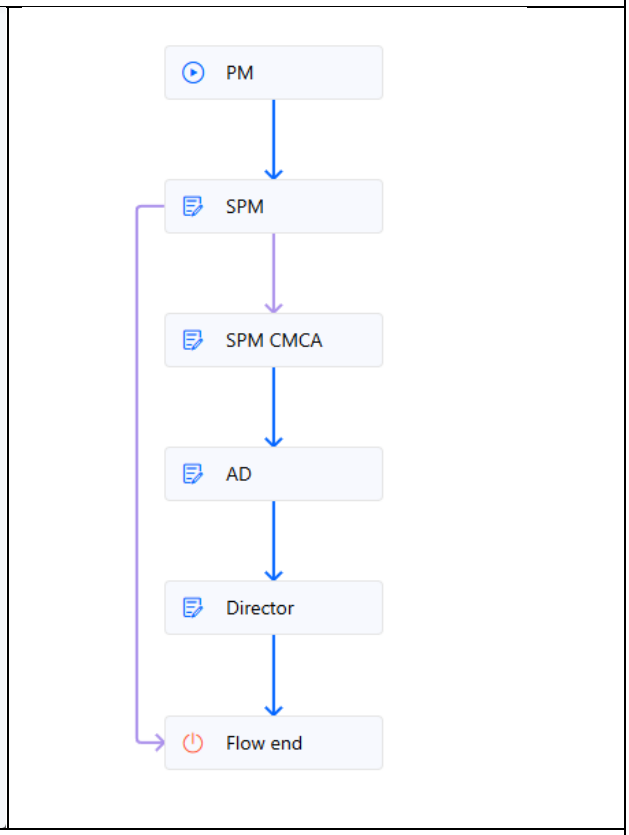
CD A1 Contract Instruction Form & Workflow



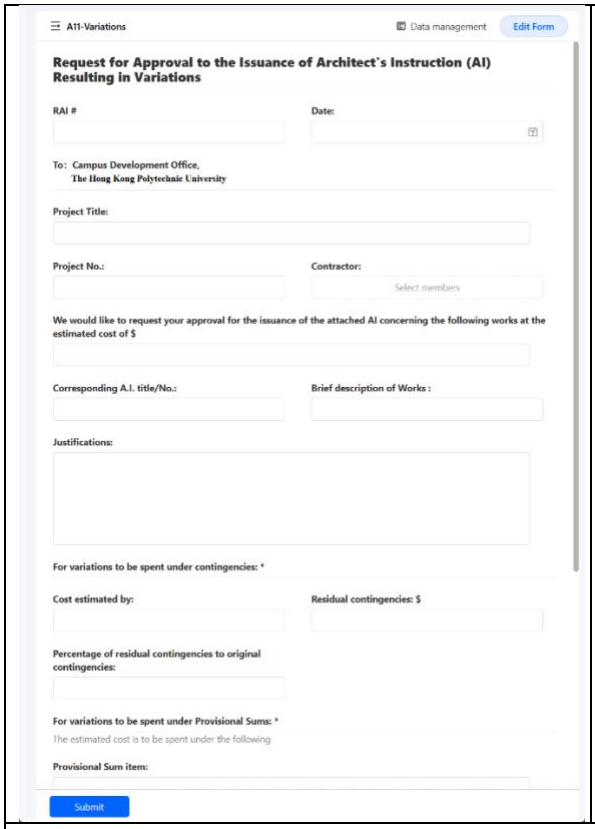


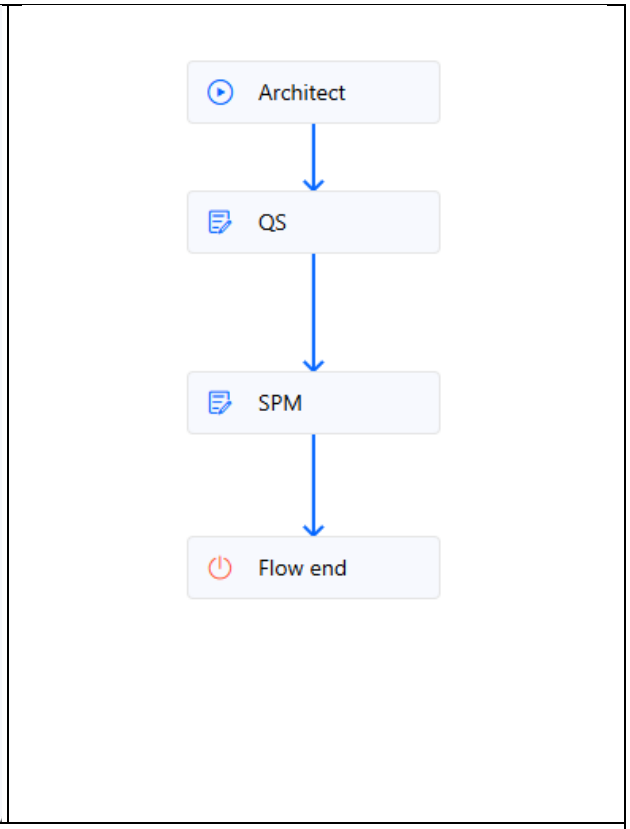
CD A6 Works Order Form & Workflow



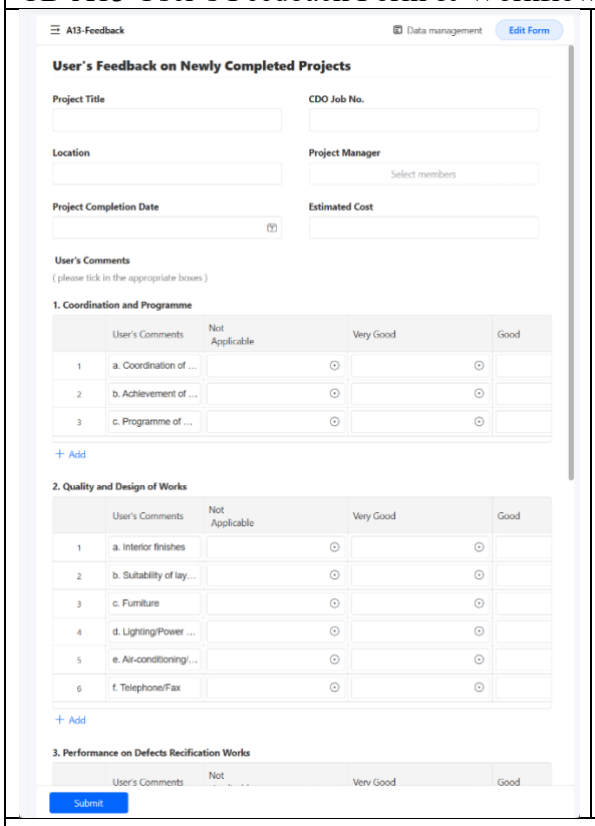


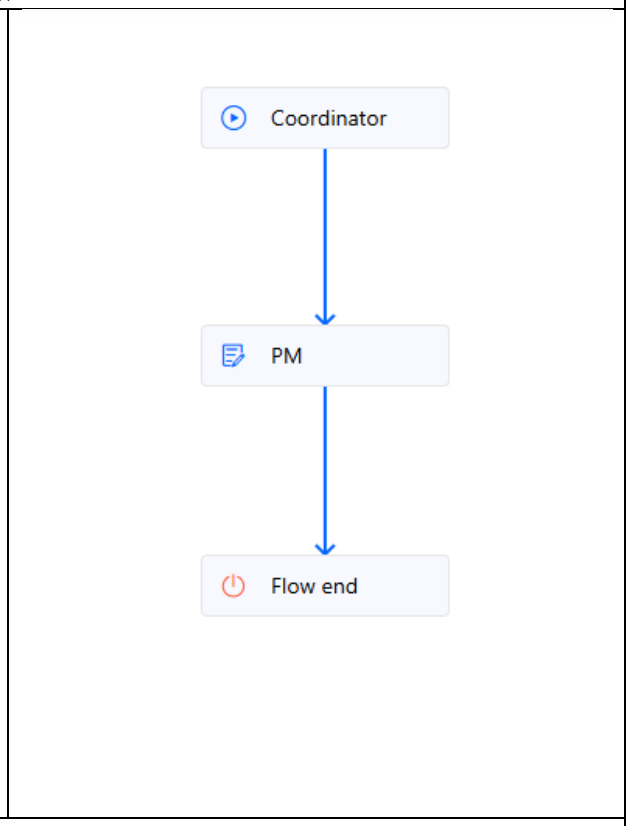
CD A11 RAI Form & Workflow



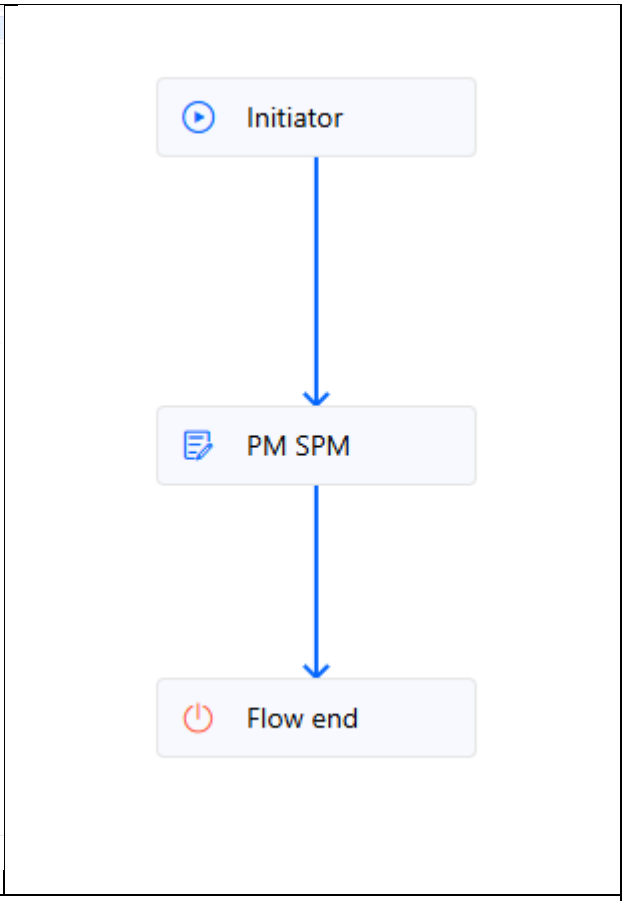


CD A13 User's Feedback Form & Workflow

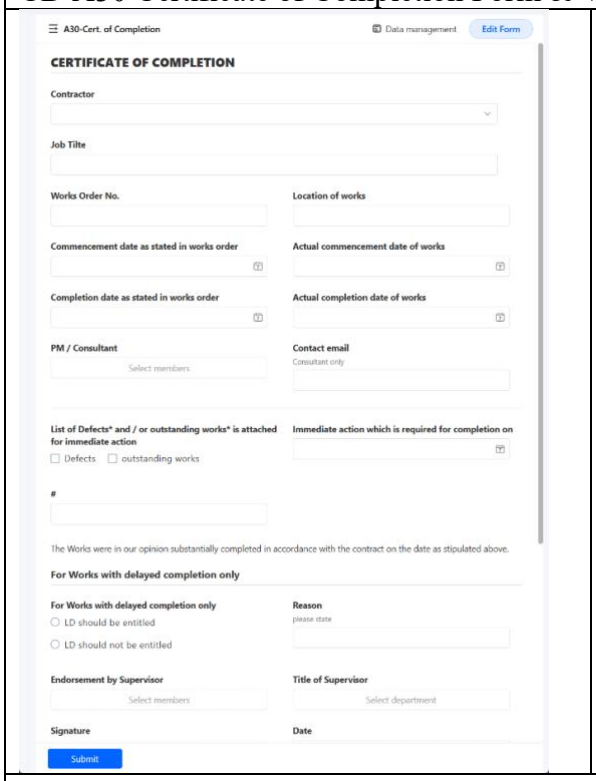


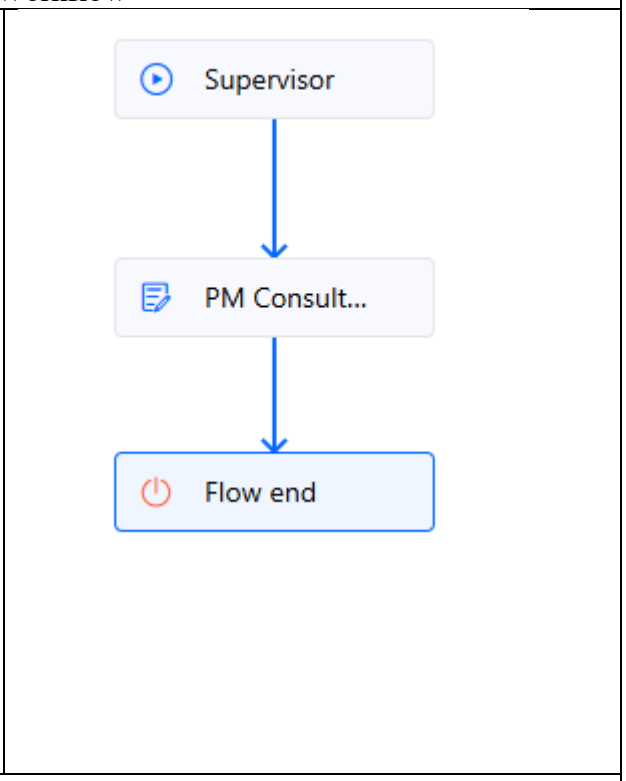


CD A28 Site Inspection Report Form & Workflow

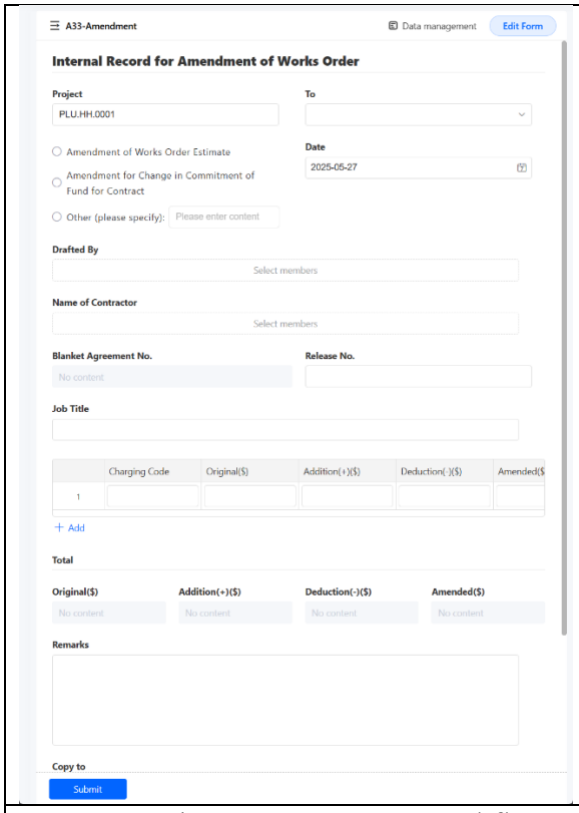


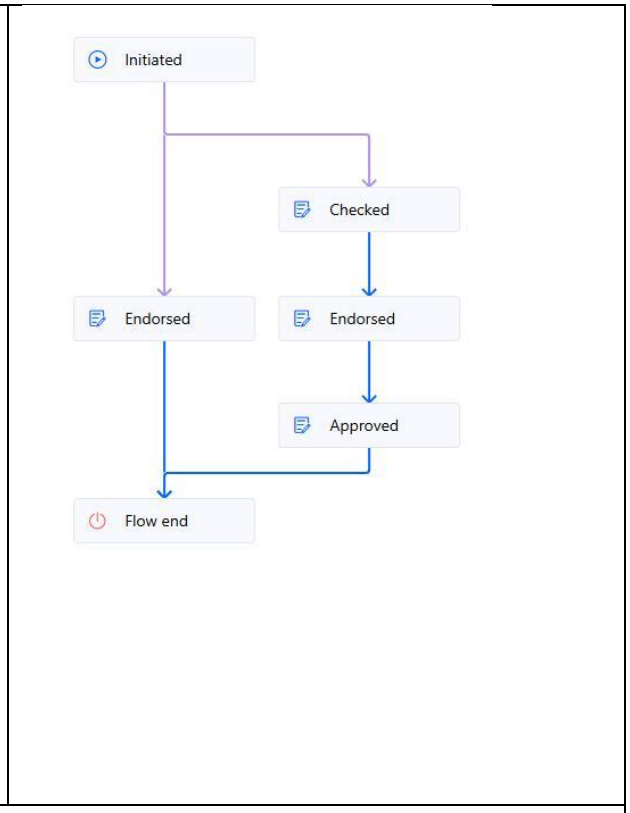
CD A30 Certificate of Completion Form & Workflow



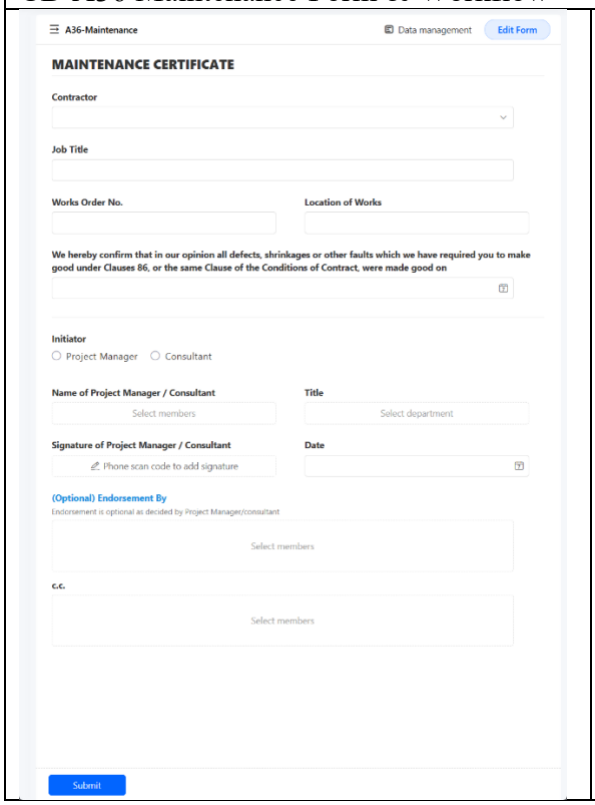


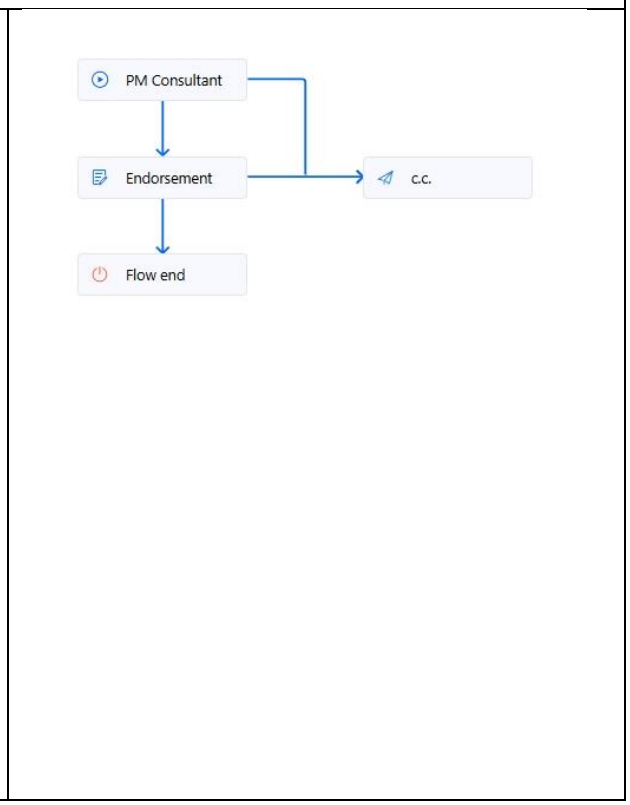
CD A33 Amendement of Works Order Form & Workflow





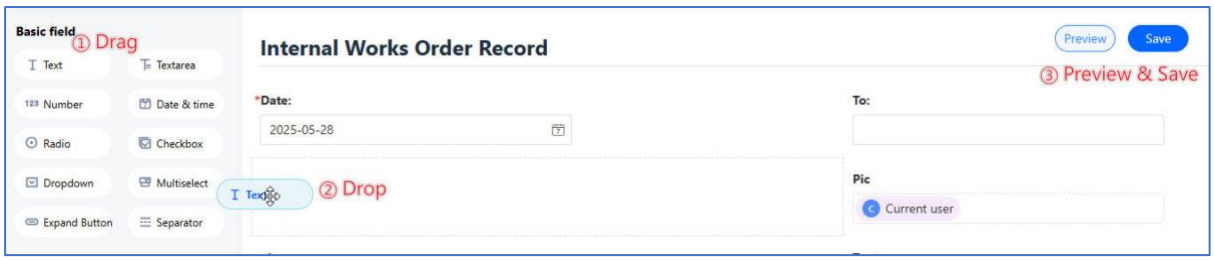
CD A36 Maintenance Form & Workflow



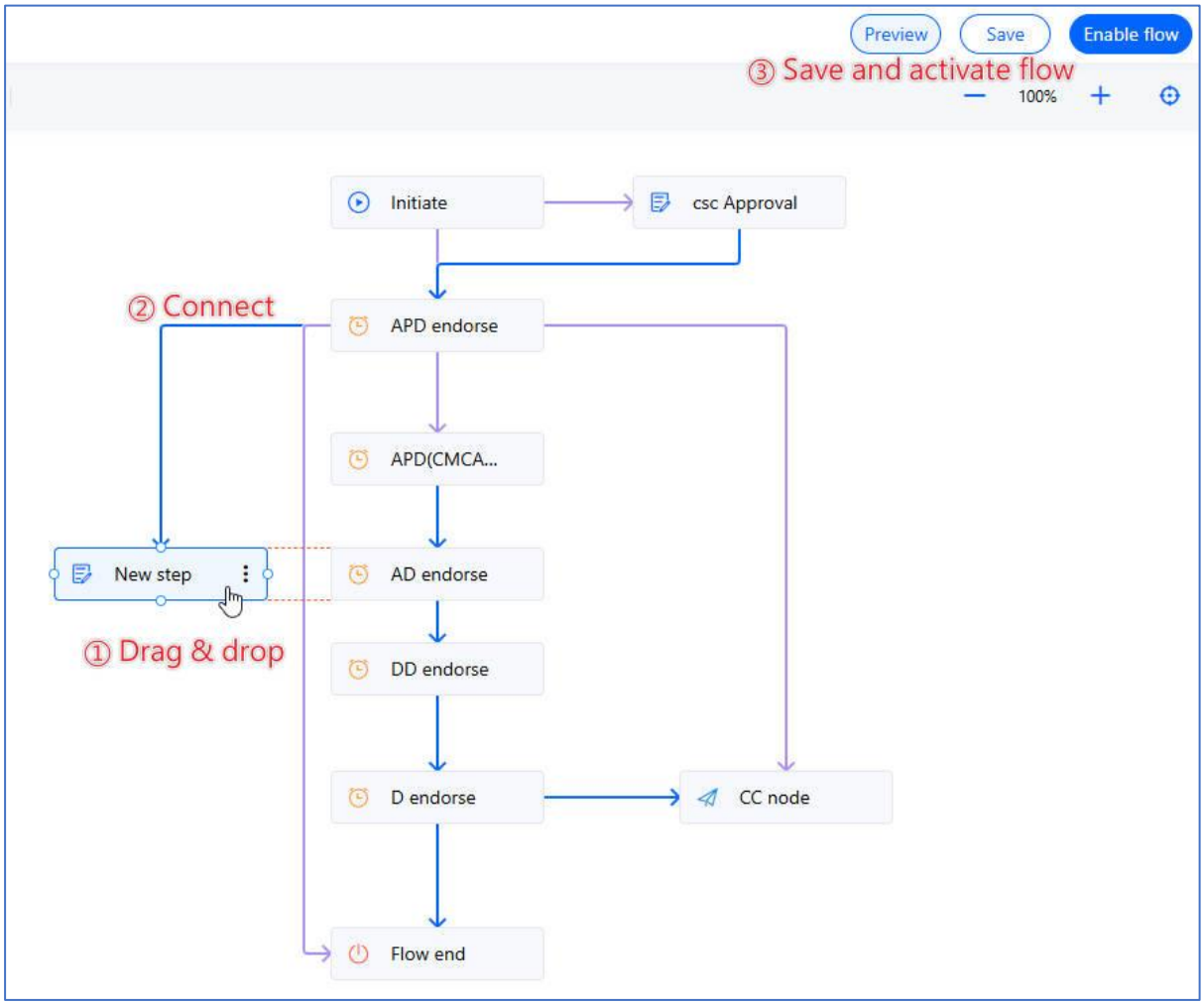


4.4.9 User Journey & Workflow Setup UX Strategy

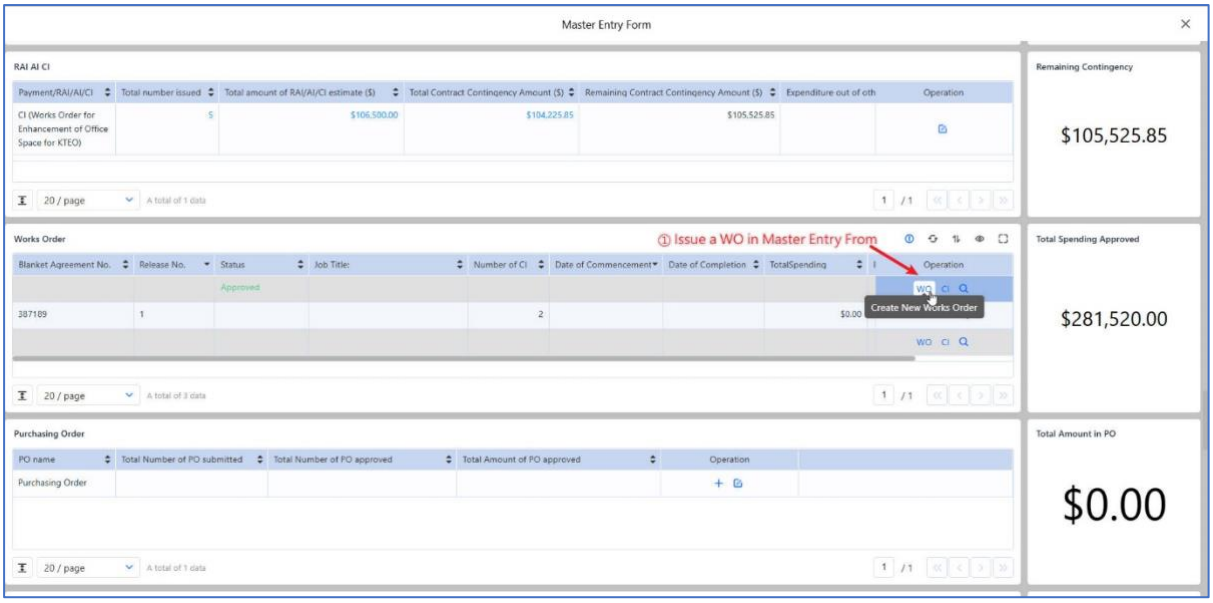
- Simplicity on form setup: 3 steps to publish a new componment in the form.



- Simplicity on workflow setup: 3 steps to adjust the workflow.

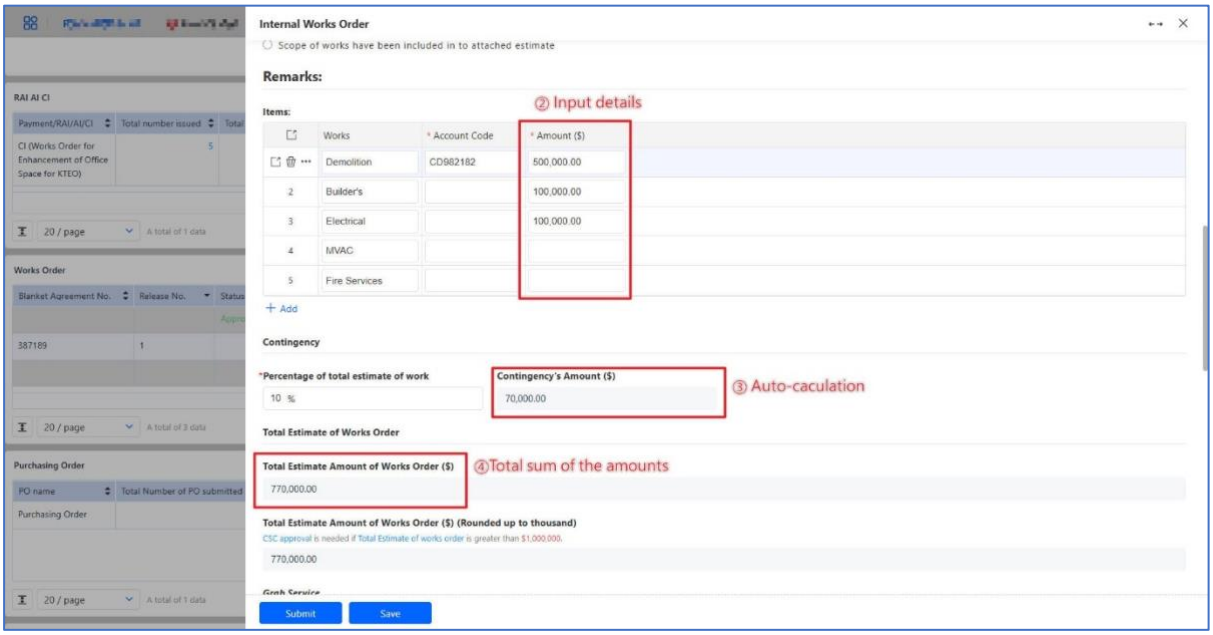


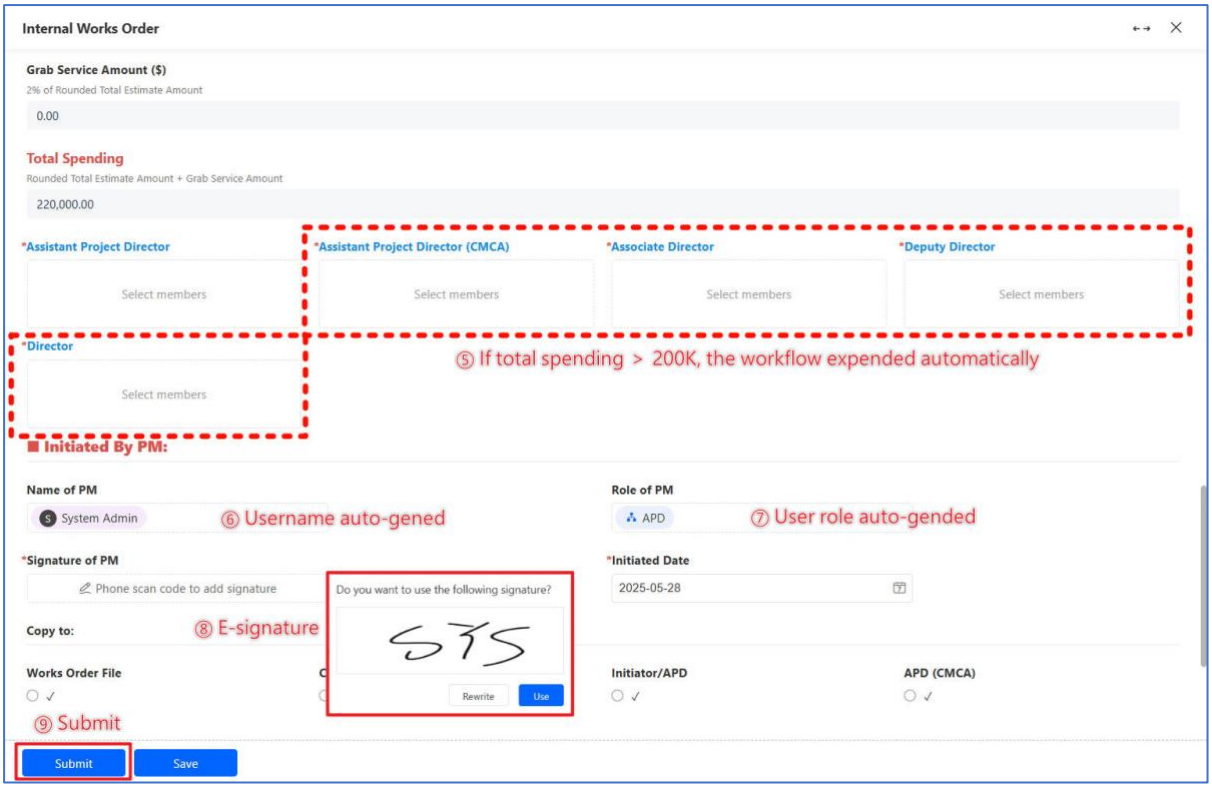
- Automated Workflow:
  - Inititate a digital form in Master Entry Form as the breakdown under a specific stage.



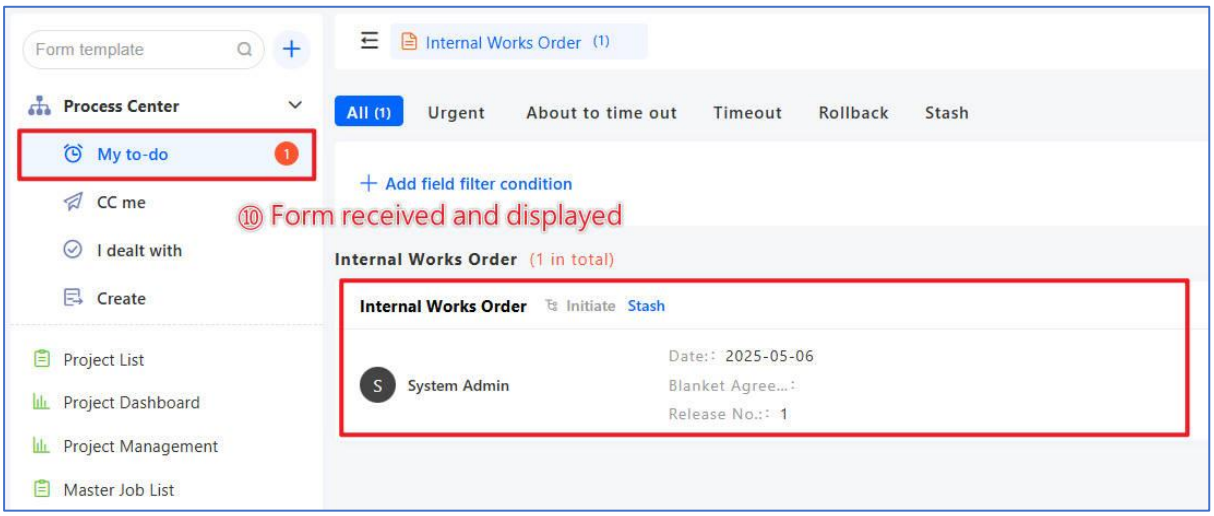
- Functions included in digital form:

a) Auto-caculation
b) Configure rules for workflow (if amount > =$200k auto-route to further approval)
c) Pre-fill known data (project title, project no., name, position)
d) Saved E-sigature (bound with personal account)

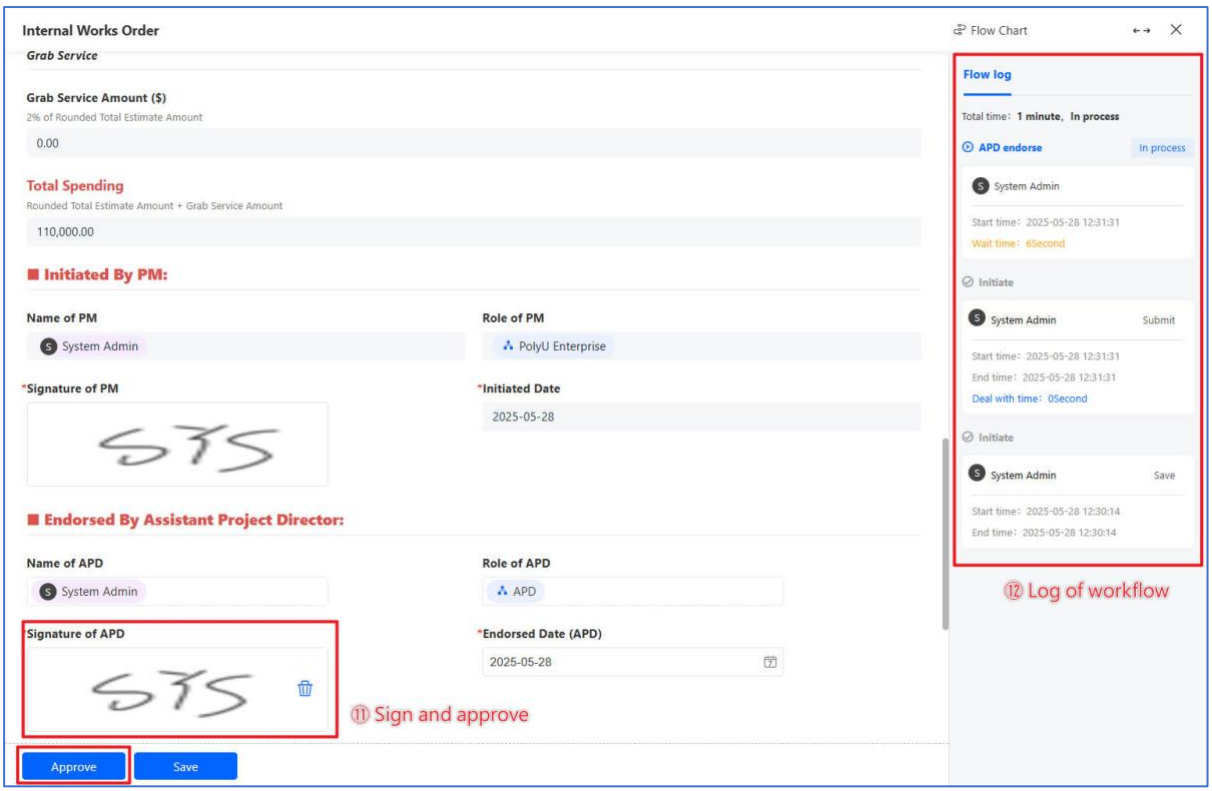




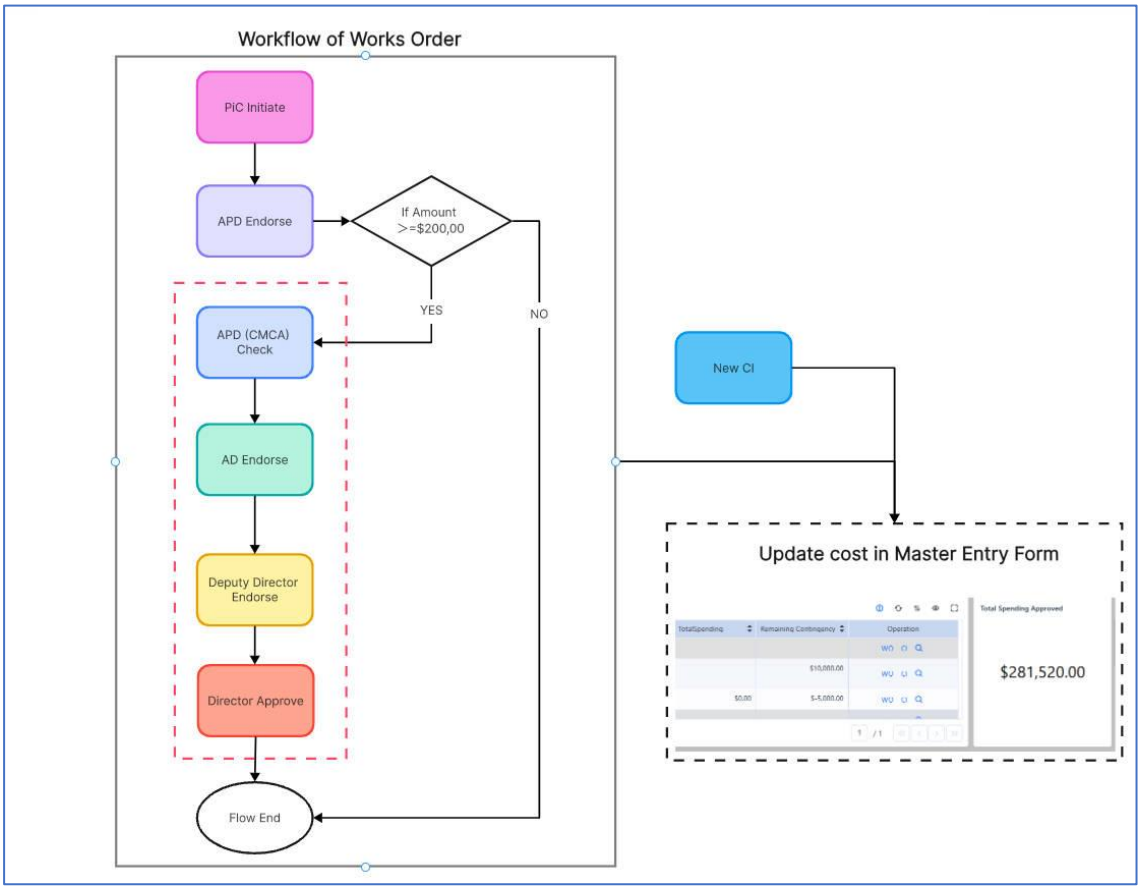
- Corrsponding action party would receive the digital forms with reminder:



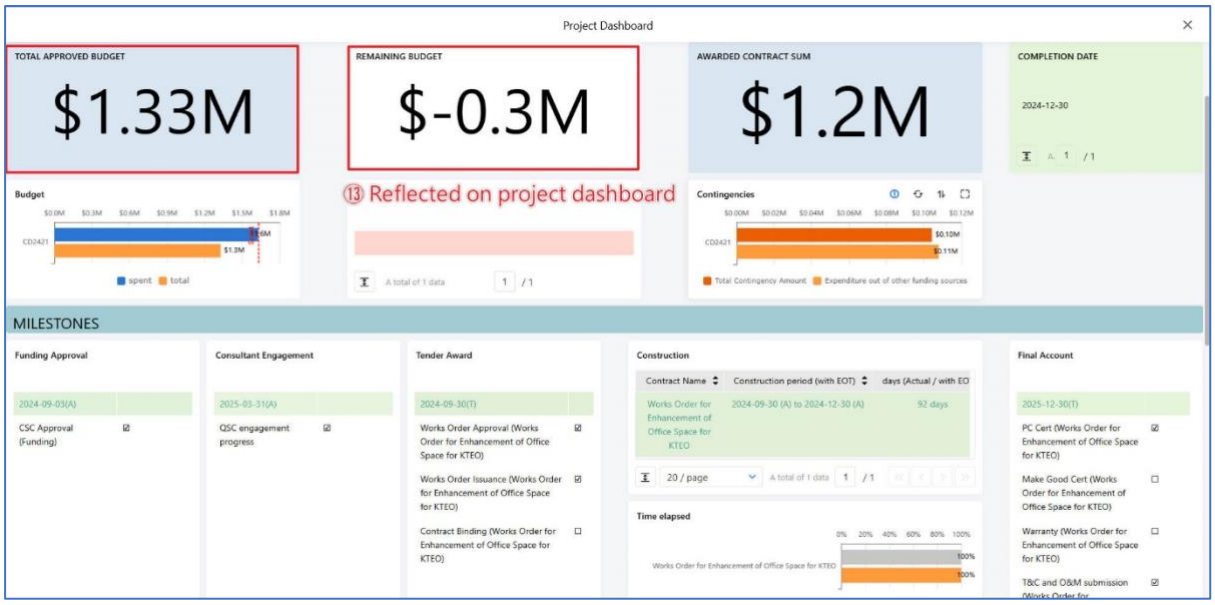
- When the application approved, the flow log recorded the approval process of the workflow:



- Flowchart example referred to Works Order: Data would be updated automatically in Master Entry Form.



- Submitted form data auto-populates to project dashboard, gloal dashboard and master job list as well:



- User journey map: From Master Entry Form to dashboards, from PiC to management grade for endorsement director

	Request	Master Entry Form	Initiate Digital Forms	Approval Procedure	Visualization
Scenario	Request creation or renewal for project information	PiC could update the project information covering stage 1 to 6 in Master Entry Form	PiC could initiate different forms in relevant stage of the project for example Works Order in Stage 5 Construction	APD, CMCA and other corrsponding parties would be reminded and asked for action according to the workflow	Data could be summarized into Master Job List in real time, and visiulized in Project Dashboard as well as Global Dashboard
Notification	N/A	Notification would be sent to remind corrsponding parties after updating project infomation	N/A	Notification by email during the workflow, and reminder before due day	Monthly Reporting
Benefit	N/A	Easy to input with checkboxes, dropdown list and other components. Also navigation bar would help to locate 6 stage quickly	Shorten the time cosumed by standradizing presettable data	Fancy to complete the endorsement with E-signature	Single source of truth as all data being in one central platform